

Starting grade	Question 1	Question 2	Extra questions
18.0	Bending vibrations	1 d.o.f. system in a force field	
18.0	Axial deformations	What is the resonance speed of the road vehicle in harmonic road way	
18.0	Torsional beams	Cross answers where I had to choose m, k and c from the ones propose. The answer was found seeing the G and the phase in the peak	
18.5	Torsional vibrations	Found m, k, c from a graph (cross question)	Caso free-free per torsional vibrations
19.5	Axial vibrations	Modal approach for DPS	1 dof stability
21.0	Axial deformation	Stability of 1 dof in force field	
21.5	Torsional vibrations	1 DOF system subjected to a force field	To find natural frequencies and modal shapes, given the boundary conditions, of the torsional vibrations
22.0	Axial vibration	Modal superimporition approach for beams	
22.0	Torsion vibration in beams	2 dof in force field	
22.0	Modal Approach dps	Railway wheelset stability	Multiple choice on evaluating the parameters from FRF 1dof sys
22.5	Modal superimposition approach	Stability of a ground vehicle in a curve	Problem on 1 DOF system subject to a force field (find equilibrium position and analyze stability)
23.0	Bending Vibration	2-DOF system Stability in a Force Field	
23.0	Modal approach for distributed parameters	Stability of 2 d.o.f in force field	1 related to frequency of pinned pinned beams, Other related to cases when K_t and $C_T > < 0$ i e stability, instability etc.
23.0	Bending vibration of beams	Yearwork	
23.5	Torsional vibration	Stability of 2 dof system	
23.5	Modal approach for dps	2 dof system in a force field	
23.5	Modal approach for dps	Exercise: Given F_x, F_y say if the 2dof system is stable	Since I answered well to both questions, prof. Zuin asked me "when does flutter instability occur in the wheelset?" which is usually asked to higher grades. The answer is in the matrices C and Kf of the system. Grade from 23.6 to 25.
24.0	Modal approach for dps	Torsional vibrations	
24.0	Modal approach for distributed parameter system	Stability for 2DOF	
24.0	Torsional beam	Stability of a 2 dof system	Spiegazione fisica della instabilita dinamica (fluttering). Ho praticamente fatto scena muta
24.0	Modal approach dps	Aeroelastic stability of rigid body in wind	
24.0	Torsional	1dof stability	Yearwork, I should add this point after that my exam n his oral questions related to yearwork fibished he showed me my grade n I saw it is as same as my written exam n I shocked(because i answered orals very well n he was pleased with my performance finally he noticed that he had wrongly considered my oral questions associated to grade 21 while my written exam grade was 24 he add 3 to 21 but after noticing his wrong he added 1 grade to me n finally 25 while i was more prepared for questions related yo my grade (2dof, modal prnciple for dps...)
24.0	Modal approach for distributed parameter systems	Stability of a ground vehicle	Cross question about pinned pinned beam excited by two equal forces but in opposite direction (one upwards and one downwards applied respectively in L/4 and 3/4L). Which mode is excited? and then Zuin gave me a sinusoidal force field like the second mode of vibration and asked me what would be the modes that will be excited. She asked me which parameter I have to change in order to transform a vehicle from oversteering to understeering one.To try to increase further the mark she asked me 3D fem (how to go from ideal cubical model to the prismatic real one) and I had no answer.
24.5	Bending vibration	Yearwork	stability of 1 d.o.f
24.5	Modal approach distributed parameter system	Railway stability	
24.5	Torsinal vibration	Stability analisis with wind force	2 cross question (FRF pin pin beam, force field)
24.5	Stability of ground vehicles	Yearwork	Question 2: in the yearwork question he asked why in the bode diagram of the FRF4 there wasn't the effect of ω_3 . The answer is that point A for ω_3 has only vertical displacement (it is seen from the mode of vibration 3) while we are looking for an horizontal displacement.
24.5	Mode superimposition for dps	Stabity of a road vehicle in a curve	Esercizio su un pinned pinned beam con due forze
25.0	Stability of a road vehicle	Yearwork	Possibili soluzioni per il punto 5 (Yearwork) = aggiungere smorzatori tra i beam più deformati nei modes of vibration
25.0	Stability of road vehicle in a curve	Modal approach for distributed parameters system	1 exercise on 1 dof stability subjected to force field. 1 extra question about bending vibration for pinned-pinned case

Starting grade	Question 1	Question 2	Extra questions
26.0	Stability of railway wheelset	Identification of modal parameters	Two short exercises on frf and conservative force fields; how to determine the critical speed for a railway system and what kind of changes we can do to increase it.
26.0	Modal identification techniques	Stability of a ground vehicle	Stability 1 dof, calculate first frequency o a beam
26.0	Stability of a 2-dof system subjected to a positional force field	3D FEM	
26.0	Identification of modal parameters	Stability of a rigid body exposed to wind	1 domanda a risposta multipla, domande di contorno sulla stabilità dei sistemi meccanici (Lyapunov, caratterizzazione della stabilità rispetto alle matrici etc.)
26.0	FEM 3D	Stability with wind	Ex:given frf of a 1dof find m,c,k
26.5	3d FEM	Wheelset stability	How do we calculate the critical velocity (scena muta, va bene lo stesso ma non alza il voto)
27.0	Stability of a railway wheelset	Torsional vibration	
27.0	Railway stability	Torsional deformation	Compute the third Frequency of bending for pinned pinned beam (multiple choice), tell (given data) if a system of 2dof and 2 forces is asy stable, simply stable, flutter inst, divergent instability (multiple choice)
27.0	Dynamics of a road vehicle	Modal approach for a distributed parameter system	
27.5	Identification of modal parameters	Yearwork	
27.5	3d fem	Modal approach	
27.5	3D fem	Stability of a railway vehicle	
27.5	Dynamic of rigid body + exemple	Railway wheelset stability	How to get critical speed
27.5	Modal identification techniques	Yearwork	
27.5	Spatial dynamics of a rigid body + example	Stability of a rigid body subject to wind action	Exercise + year work in particular q.5
28.0	3d fem	Modal approach for flexible sys	
28.0	3d fem	Stability of a railway wheelset	How to compute the critical velocity of railway (answer: when the real part of at least one of the lambdas becomes positive)
28.0	Modal identification approach	Yearwork	Domanda sullo yearwork (mai fatta prima da come ha detto la Zuin): se volessimo usare il metodo FE per un sistema fortemente non lineare, che metodo dovremmo usare? Risp: metodi numerici come RK
28.0	Modal identification techniques	Yearwork	Given f1=5 f2=20 f3=80, choose T and fs
28.0	Modal identification techniques	Yearwork	How to apply FEM to highly nonlinear systems: numerical integration methods. Flutter instability for real airplane wings: given by the coupling of bending and torrio al motion.
28.0	Modal parameter Identification	3D rigid body+example	Yearwork, Q: why there is no peak for 3rd nat freq, plot: horizontal force at point A, output:horizontal displacement of point A. Answer: in deformed structure related to 3rd nat freq, we had no horizontal disp for point A
28.0	Dynamics of a rigid body (+ example)	Solid FEM	Exercise about FRF and equation of motion of a 1 dof system
28.5	Railway wheelset stability	Modal parameters identification	Difference with high speed trains suspensions, detailed description of impulsive excitation for parameter identification.
28.5	3D fem	Yearwork	
28.5	Identification of modal parameters	Yearwork	
28.5	3D FEM	Yearwork	
28.5	3D body (with example)	Model identification techniques	Question about distributed parameter system and impulsive periodic force (Fourier series)
28.5	3D body with example	Stability of road vehicle in a curve	Yearwork
29.0	Identification of Modal Parameters	Aeroelastic Stability of a 2-DOF Sys	
29.0	Modal identification	Stability to wind	
29.0	Rigid Body in space and example	3D fem	Problems of distortion, principal axes in rigid body example
29.0	Modal identification	Yearwork	2 mini exercises: stability of a 1dof system and identification of parameters from a graph
29.0	3D fem	Yearwork	
29.0	Model identification Parameters	Yearwork	
29.0	Model identification	Yearwork	
29.0	3D FEM	Yearwork	Perché la trasformazione nel mapping deve essere biunivoca, caso degenero due nodi coincidenti
29.5	Stability of a wheelset	Modal parameters identification	

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29.5	stability of a railway wheelset	modal identification techniques	2 short multiple choice questions, conclusion about the reaction forces in dyn. of a r.b. in space, how to increase the critical speed in a railway sys., what changes in the design of an high speed train (we increase the stiffness of the wheels to increase the critical velocity and operate at higher speeds, but the curves must be large cause the wheels are not that flexible and cannot abruptly rotate since they're stiffer than usual), hunting motion, Lyapunov sys (linear-non linear relationship)
29.5	Dynamic of a rigid body in space, including example done during lectures	3D FEM	
29.5	Modal Identification Techniques	Dynamics of a Rigid Body in Space + example	1) Exercise about a rigid body in space subject to a force with three components 2) Exercise about force field and verify flutter instability
30.0	Dynamics rigid body 3D	Yearwork	
30.0	Modal identification techniques	Aeroelastic stability of a body in wind	How to infer stability/instability properties of a system given a FEM model? -> very high number of dof -> use modal decomposition and consider modal coordinates related to frequencies of interest
30.0	Modal identification	Yearwork	
30.0	Modal Identification Techniques	Yearwork	Suppose that you lead an experiment on a system which has the following natural frequencies: 5 Hz 20 Hz and 80 Hz. Define some reasonable values for the data acquisition time T and the sampling frequency fs. Answer: fs on the order of 300 Hz or more and T = ranging from 1 to 10s is fine.
30.0	Model identification	3D rigid body+example	How to move the axis of rotation so that it coincides with the com G? (Add concentrated mass to the system). If a vehicle vibrates due to unbalanced mass in the wheels for a V=100km/H but I notice that decreasing/increasing the V makes the vibrations stop, what could be the reason? (I am stepping out of the resonant frequency of the system)
30.0	Modal identification techniques	Yearwork discussion	Flutter instability, 3d body example
30.0	Spatial Dynamics of a Rigid Body	Stability of the wheelset	
30.0	Dynamics of a rigid body + example (rotation around an axis)	Modal parameters identification	
30.0	Dynamic of the 3D Rigid body + example	Yearwork	
Higher	Stability of Train Wheelset	Yearwork	